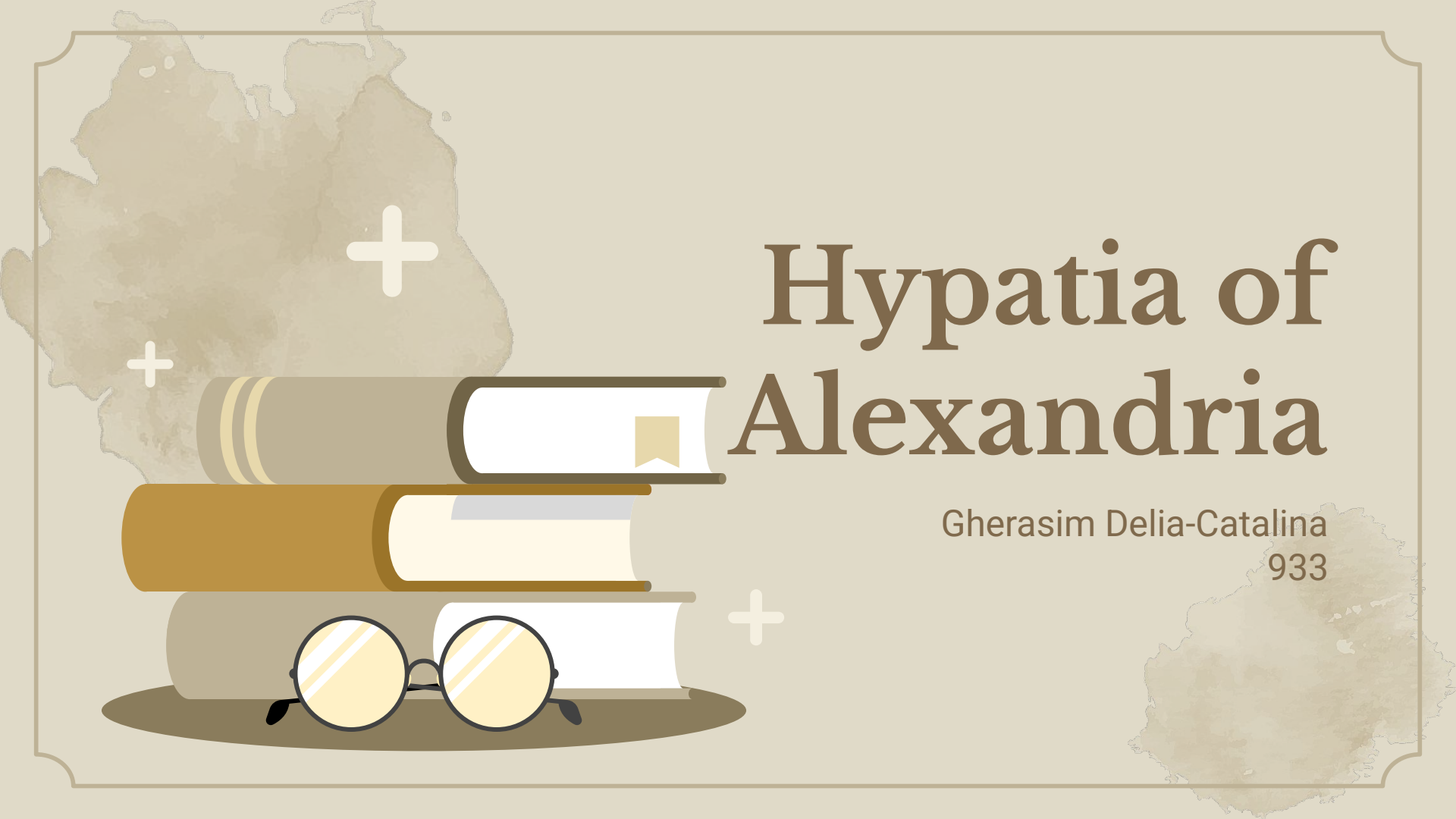




# Hypatia of Alexandria

Gherasim Delia-Catalina

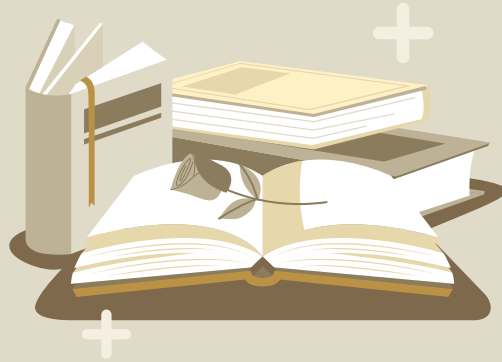
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# Introduction

Hypatia of Alexandria was one of the few female mathematicians and philosophers.

Little is known of her early life, but in a time when women were not allowed a formal education, she stood out becoming a respected teacher and thinker. In many ways, she paved the way for female mathematicians.

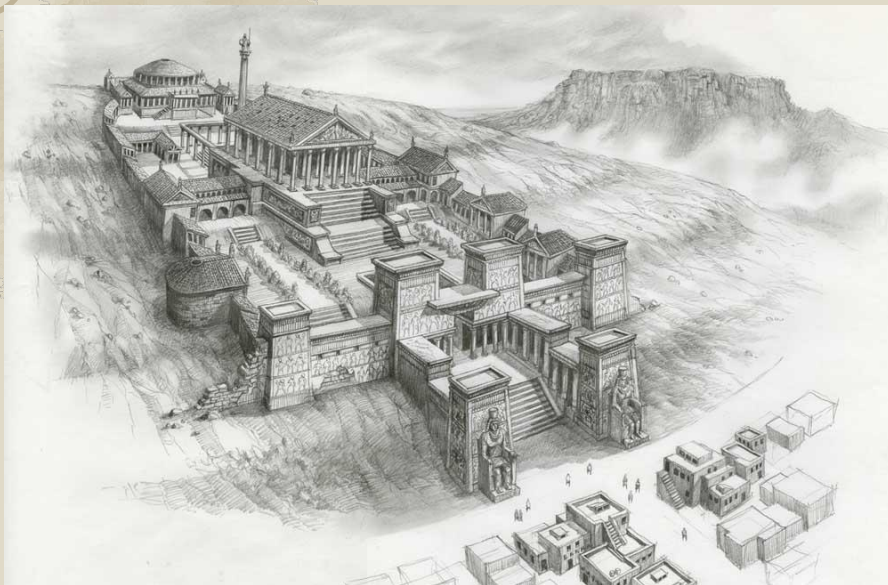
Her notes on math and astronomy helped modern mathematicians and astronomers come up with advanced theories that are still used today. **Her death marked the end of Greek science.**



# Historical Context

During 300 BC the Egyptian ruler Ptolemy I ordered the construction of the largest scientific establishment in the world: a higher education institution and a library. The school flourished, preserving Greek knowledge and receiving students from all over the world. In the Alexandria School there were no examinations, and no titles were awarded. In the medicine field, the fact of having studied there was considered sufficient to be able to practice.





**Alexandria, the city founded by Alexander the Great facilitated an intellectual center of the ancient world, influenced by Greek, Egyptian, Roman and Jewish minds.** By the Roman period the ethnic and religious diversity contributed to conditions for women in Alexandria that were more liberal than for women elsewhere in the ancient world.

**While girls were not generally welcomed into schools of rhetoric and philosophy, there is some evidence that a few allowed, and even welcomed them.** Some girls and young women probably learned philosophy by listening to conversations of men or by male relatives who were teachers, more common in cities known for excellent education opportunities like Alexandria. As an example, in Athens, Agnodice or Agnodike, a woman who pretended to be a man to study at the School of Alexandria with Herophilus, successfully practiced as a gynecologist and was brought to trial for "pretending to be a man to practice medicine".

The library suffered through a fire in 48BC, destructions due to war in 272 and in 391 ravaging by Christians who declared it a pagan place.

*“Alexandrian schools had long integrated pagan and Christian students without incident and throughout the beginning of the third century students of the Alexandrian Christian community maintained close ties to pagan teachers. For the cultivated Alexandrians, pagan philosophy and Christianity were “perfectly compatible” because the combination created the building blocks of philosophical study.”*

Unfortunately, by the fourth century, the city endured a significant change: the growth of Christianity threatening social unity. In 391, Emperor Theodosius outlawed paganism and encouraged the destruction of pagan temples, beginning a wave of violence against pagans in the Roman empire.

*“And thus the stage was set by the late fourth century for a veritable war of ideas waged by letters, speeches, and riots. Many adherents to the Greco-Roman tradition clung valiantly to their beliefs even as the new religion won over hundreds of thousands of new converts.” (1)*





### 03 Life, education and social status

The exact date of Hypatia's birth is unknown but is believed to be between 355-370 A.D. Born in Alexandria, the only child of mathematician and astrologer Theon, the director of the University of Alexandria. The name of Hypatia's mother is not known.

It is noted that her father "*must have had lofty hopes for his daughter because he gave her the name Hypatia, which means „the highest“.*" Her father taught her everything he knew about mathematics, and she equaled him in nearly all subjects and surpassed him in astronomy.

She learned Neoplatonism philosophy from a teacher other than her father. Since Theon probably conducted Hypatia's education in the Museum, Hypatia would have been exposed to a wide variety of scholars, among them, Neoplatonists, Christians, and Gnostics. Her access to the Museum also means she could have taught herself philosophy by reading the scrolls stored there.



When her father retired, Hypatia took over his role as teacher. Her wisdom attracted many students from privileged backgrounds. Given the expense of an education, the fact that wealthy and sometimes powerful families sent their sons to Hypatia's school speaks to her renown as a teacher.

Hypatia's most famous student was Synesius of Cyrene, an important figure in Eastern Christianity. Although he only remained in Alexandria until about 395, he and Hypatia corresponded until his death in 412. Many of his letters still exist, though none of her replies do. What is clear from his letters to Hypatia is that he held her in great esteem.



As a woman of Greco-Egyptian decent, Hypatia's public work was supported by Egyptian, Greek, and Roman legal and social customs that enabled her to lecture in public and private, administer her own school, and advise high-level political leaders. Hypatia was part of the culturally Greek intellectual elite. It is important to highlight that Alexandria under Roman rule was an unusually liberated place for women. A woman of mixed ethnicity such as Hypatia could depend on law and custom for greater empowerment in Alexandria than elsewhere in the ancient Mediterranean. For example, Hypatia did not appear in public with a guardian, which was a common African custom.





In 415, as she was riding home a group of monks dragged her from her chariot, stripped her clothes and killed her. The Church historian Socrates Scholasticus writes that the Christians blamed her for the falling out between the proud Patriarch of the Alexandrian Church Cyril and the Roman prefect Orestes, whom she was close with.

# 04 Mathematical<sup>+</sup> contribution<sup>+</sup>

The Suda Lexicon, the tenth-century Byzantine encyclopaedia, attributes to her 3 works: a Commentary on Diophantus, one on the astronomical Canon and a commentary on Apollonius's Conics. Commentaries are what we would refer to as editions, with the mention that the commentator would often add new material.



# Theories



## APOLLONIUS'S CONICS

The Conics are made up of 8 books, and only 7 survived via Eutonicius's Commentary or the Arabic. The final one, as well as Hypatia's Commentary, are lost



## THE ASTRONOMICAL CANON

Theon, her father, wrote commentaries on these 13 books. Book Three contains the inscription "in the recension of my philosopher-daughter Hypatia."



## DIOPHANTUS' ARITHMETIC

The most likely contributions of Hypatia are 2 "Student exercises" at the start of Book 11, asking for the solution to "the pair of simultaneous equations"

1.  $x - y = a$   
 $x^2 - y^2 = (x - y) + b$
2.  $x - y = a$   
 $x^2 - y^2 = m(x - y) + b$

# Inventions

## The Astrolabe

**Device replicating the motion of stars.**

The theory of the astrolabe and the details of its construction were passed down from Ptolemy (author of the Astrological Canon, commented by Theon) to Theon and then his daughter Hypatia, and finally to her student Synesius.

Synesius states in a letter to Paianos that he designed the astrolabe with help from Hypatia and crafted it with the best silversmiths



## The Hydroscope

In Letter 15 Synesius writes to Hypatia: "I am reduced to this, that I have to have a hydroscope." He then described the specifications of the device needed. The translation seems inappropriate as "hydroscope" implies a clepsydra, and the device described illustrated a hydrometer, used to measure the weight of water.

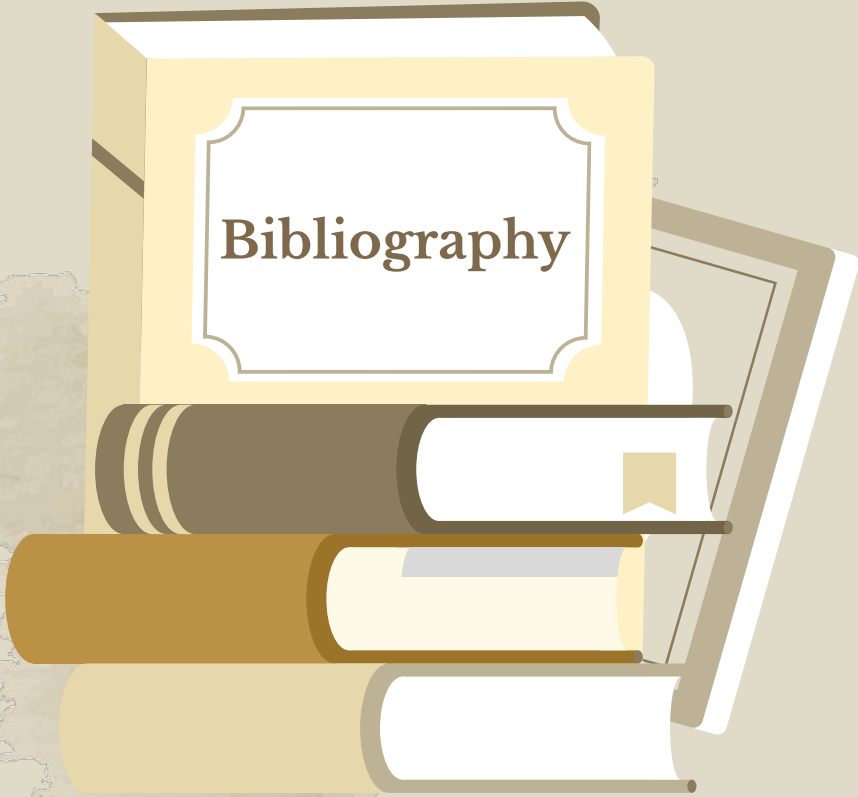


## 05 Conclusion

The legend of Hypatia, martyr of philosophy, has become inextricably linked to the wider story of the decline of philosophy in the classical world. Her grizzly murder is like a narrative climax that ends with the victory of Christianity over paganism and of orthodox thinking over debate and discussion. She is considered the last Greek thinker.







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